

# 1 Retrieval vs. ranking vs. outputs (Q&A)

## 1.1 Q: Is this a retrieval model?

**A:** No: the provided scripts score already-observed (user, video) interaction records from a dataset loader and do not implement large-scale candidate generation (ANN/top- $K$  retrieval) over a corpus.

## 1.2 Q: Is this a ranking model?

**A:** Not in the training-objective sense: the training loops use pointwise losses (likelihood/regression-style) per example rather than pairwise/listwise ranking losses over candidate sets.

## 1.3 Q: What is the final output used for evaluation?

**A:** Each model produces a scalar watch-time prediction per example for evaluation (e.g. EGMN uses the mixture mean; other baselines output scalars or reconstruct scalars from intermediate representations).

## 1.4 Q: What does EGMN output internally?

**A:** EGMN outputs mixture-parameter tensors  $\pi$  logits,  $\lambda$ , and  $(\mu, \sigma)$  for multiple Gaussian components, which define the mixture distribution and its mean prediction.

## 1.5 Q: What does the EGMN evaluation loop look like (schematically)?

**A:**

```
with torch.no_grad():
    for features, label in dataloaders["test"]:
        y = model.predict(features) # scalar per example
```

## 1.6 Q: What are typical output shapes for some baselines?

**A:**

```
# Wide&Deep: scalar
p = wide_and_deep(features) # [batch]

# CREAD: per-bucket/head probabilities
bucket_probs = cread(features) # [batch, M]

# D2Q: scalar quantile (later mapped back to value)
q = d2q(features) # [batch]
```

## 1.7 Q: If training is pointwise, why report XAUC?

**A:** XAUC is a ranking-style evaluation metric computed by sorting examples by predicted score and measuring label-order agreement, but it does not change the fact that the provided training scripts optimize pointwise objectives.